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Review

Therapeutic potential of apelin and Elabela in cardiovascular disease

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ABSTRACT

Apelin and Elabela (Ela) are peptides encoded by APLN and APELA, respectively, which act on their receptor APJ and play crucial roles in the body. Recent research has shown that they not only have important effects on the endocrine system, but also promote vascular development and maintain the homeostasis of myocardial cells. From a molecular biology perspective, we explored the roles of Ela and apelin in the cardiovascular system and summarized the mechanisms of apelin-APJ signaling in the progression of myocardial infarction, ischemia-reperfusion injury, atherosclerosis, pulmonary arterial hypertension, preeclampsia, and congenital heart disease. Evidences indicated that apelin and Ela play important roles in cardiovascular diseases, and there are many studies focused on developing apelin, Ela, and their analogues for clinical treatments. However, the literature on the therapeutic potential of apelin, Ela and their analogues and other APJ agonists in the cardiovascular system is still limited. This review summarized the regulatory pathways of apelin/ELA-APJ axis in cardiovascular function and cardiovascular-related diseases, and the therapeutic effects of their analogues in cardiovascular diseases were also included.

1. Introduction

In light of the changing modern lifestyle, cardiovascular disease has emerged as a major challenge confronting the medical community. According to recent studies, heart disease remains the leading cause of death among residents of the United States, with 690,882 deaths recorded in 2020, and an increasing trend observed from 2015 to 2020 [1]. Therefore, investigating the etiology of cardiovascular disease is of paramount importance. The cardiovascular system is a highly regulated entity, with apelin and Elabela emerging as a critical cardiovascular polypeptide that has received significant attention over the past few decades. Apelin and Elabela (Ela) are polypeptides encoded by the APLN and APELA genes, respectively. They can be secreted into the bloodstream by adipocytes and various other cells to regulate the cardiovascular system, endocrine system, and other systems, thereby maintaining homeostasis in the body. In 1998, Tatemoto and colleagues discovered apelin, a new ligand that binds to the G protein-coupled receptor APJ [2]. This receptor was accidentally identified by O'Dowd and colleagues in 1993 while attempting to clone the angiotensin receptor gene via PCR [3]. Although APJ shares 54% similarity with the angiotensin receptor, apelin does not bind to AT as a ligand, suggesting that apelin possesses distinct functions from angiotensin [3]. In recent years, other ligands such as Elabela and Tollder that bind to APJ have been discovered, producing corresponding physiological effects [4].

The apelin and Ela have demonstrated significant efficacy in treating atherosclerosis, heart failure, hypertension, pulmonary hypertension, and certain congenital vascular diseases, indicating apelin's tremendous therapeutic potential in the treatment of cardiovascular diseases [5–10]. According research published by Stiller B et.al reveals that 1% of the population suffers from congenital heart disease, and this number is on the rise, with no effective treatments or preventative measures currently available [11,12]. Apelin plays a pivotal role in the embryonic development and migration of cardiac progenitor cells, and some researchers believe that apelin may be useful in treating some congenital heart diseases [13]. However, few recent reviews have examined the therapeutic effect of apelin, and there is a lack of systematic discussion and integration. Therefore, this paper aims to review the pathological and therapeutic effects of apelin and Elabela in congenital and adult cardiovascular diseases, offering valuable insights for corresponding clinical treatment.

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2. Elements, physiological function and signaling pathway of Apelin/Elabela-APJ axis

Apelin and Elabela (ELA) are both proteins encoded by the APLN and APELA genes, respectively [2]. They can bind to the APJ receptor, which is encoded by the APLNR gene. APJ is a 7-transmembrane receptor belonging to the GPCR (G-protein coupled receptor) family. When apelin or ELA binds to the APJ receptor, which is coupled to various G proteins, different functions can be exerted in different cells and tissues. Apelin, ELA, APJ, and their downstream signaling molecules collectively form the apelin/Elabela-APJ axis, which plays a crucial role in cardiovascular diseases.

The apelin/Elabela-APJ axis is a cascade signaling pathway composed of multiple components. Upon binding of apelin or ELA to the GPCR receptor APJ, G proteins are activated. Within this axis, various types of G proteins can be activated, including Gas, Gai, Gq/11 [4], and others. Once these G proteins are activated, they can trigger multiple downstream signaling pathways such as MAPK/ERK, PI3K/AKT, PLC/IP3, ultimately leading to gene expression activation or acting on other organelles to exert their effects [8]. Additionally, studies have shown that apelin can increase the expression of ACE2, which in turn leads to the degradation of apelin and a decrease in its activity. This indicates that the apelin/ELA-APJ axis can also promote the degradation of apelin through increased ACE2 expression to avoid sustained activation of APJ [14]. Furthermore, APJ, as a GPCR receptor, can also be downregulated through β-arrestin-mediated receptor desensitization. This evidence indicates that the apelin/Elabela-APJ axis is also regulated by negative feedback, ensuring that apelin and Elabela can exert physiological effects of appropriate intensity in the body to maintain cardiovascular system homeostasis. In this section, we describe the structure and biological functions of Apelin, Ela, APJ and their downstream signaling.

2.1. Apelin structure and function

Apelin is a peptide hormone encoded by the APLN gene located on Xq25–26 [2,15]. The pre-protein form of apelin contains 77 amino acids and is processed by endopeptidases to generate proapelin, which consists of 55 amino acids [16]. Proapelin is further processed by proprotein convertase subtilisin/kexin 3 (PSCK3) to produce shorter forms of apelin, including apelin 36, apelin-17 and apelin-13[17,18] (Fig. 1) .

Interestingly, recent studies have demonstrated that PSCK3 can directly cleave proapelin to remove unwanted peptide chains from the N-terminal, leaving only the core residues at the C-terminal to form Apelin-13 [17]. Notably, the C-terminal region of apelin is highly conserved, with twelve residues being critical for its biological function [19]. It is noteworthy to mention that both apelin-13 and apelin-17 can act as substrates for angiotensin-converting enzyme 2 (ACE2) [20]. Following ACE2 cleavage, the C-terminal phenylalanine is removed, resulting in the formation of apelin-12 and apelin-16, respectively [20]. It has been demonstrated that both apelin-12 and apelin-16 exhibit a significant decrease in biological activity [20]. In the study conducted by Janet J. et al., ACE2-processed apelin-13 and apelin-17 exhibited a reduced ability to promote the production of nitric oxide (NO) in HUVEC, as well as insufficient AKT phosphorylation in cardiomyocytes, ultimately leading to a diminished ability to protect the heart and blood vessels. Therefore, it is plausible that the C-terminal phenylalanine of apelin peptides in the apelin family is crucial for activating the receptor and producing the corresponding effects. The cleavage of apelin by ACE2 may be considered the initial step in the degradation of apelin. If related drugs are to be developed, it is crucial to note the protection of the C-terminal phenylalanine of apelin.

In addition to Apelin-13, pyroglutamate-modified Apelin-13, known as Pyr-Apelin 13 has been identified. Both Apelin-13 and Pyr-Apelin 13 exhibit high affinity for the apelin receptor and are highly expressed in the cardiovascular system, where they play important roles in vasodilation, cardioprotection, and the promotion of vascular endothelial cell proliferation and migration [21]. Also, apelin-13 has an antithrombotic effect by blocking platelet function [22]. Specifically, treatment with apelin-13 inhibits the activity of integrins, including αIIbβ3 and α2β1, which leads to the reduction of platelet adhesion to fibrinogen and collagen, so they predicted apelin could be a therapeutic target to treat thrombosis [22]. Furthermore, based on previous evidence, apelin has been shown to exert positive inotropic effects in the treatment of heart failure, without causing myocardial hypertrophy [23]. In summary, apelin plays a crucial role in the cardiovascular system and holds great potential for therapeutic applications.

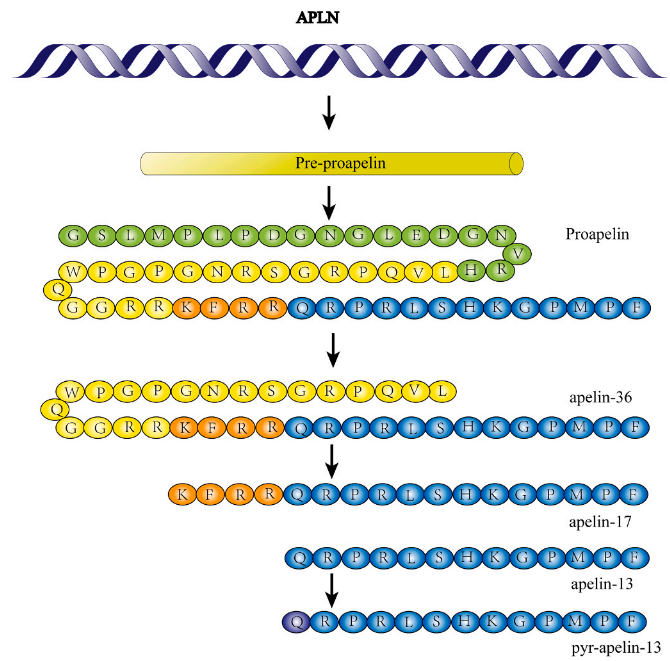

Fig. 1. The pathway of formation of apelin and product. The APLN gene is translated into pre-proapelin, which undergoes hydrolysis by multiple enzymes to generate apelin-36, 17, and 13. The N-terminal glutamine of apelin-13 can be modified into pyroglutamylate form to produce pyr-apelin-13. The modified amino acid is highlighted by purple.

2.2. Discovery and function of elabela

After the discovery of apelin, researchers conducted extensive studies on the expression sites of apelin and the time when apelin is expressed in the human body. APJ is expressed earlier in embryonic development than apelin [24]. APJ is expressed before the occurrence of the primitive gut tube, but apelin begins to be expressed in the middle and late stages of primitive gut tube formation. Some studies have also shown that APJ-deficient mice were not born in the expected Mendelian ratio, and many showed cardiovascular developmental defects, while apelin-deficient mice survived, reproduced, and developed normally [13,25]. This suggests that APJ has other undiscovered ligands.

Until 2013, the second receptor of APJ was discovered in zebrafish embryos and named Elabela [4]. Elabela is encoded by a gene on chromosome 1 and is processed and cleaved by enzymes in the endoplasmic reticulum to form elabela-54, elabela-32, elabela-22, and finally the evolutionarily conserved 11-peptide elabela-11 through furin cleavage [26].

Elabela is widely expressed in human embryonic stem cells and primitive gut tube, and then plays a regulatory role in cell migration and differentiation in the endoderm and mesoderm [27,28]. It is particularly important for myocardium lineage migration to APLM and cardiomyocyte migration to the first and second heart fields [13,25]. Severe cardiac developmental defects were observed in elabela$^{\Delta/\Delta}$ mouse embryos, and although blood vessels developed, the sprouting of blood vessels was severely damaged, ultimately leading to miscarriage [8]. The proper development of the cardiovascular system is critical for lower vertebrates, and Ela plays a crucial role in this process [4,29]. Lack of Ela in zebrafish embryos results in impaired migration of the cardiac progenitor cells, leading to severe cardiac malformations [4]. Additionally, research suggests that Ela can regulate fetal development, maternal blood pressure, renal function, and normal placental development during pregnancy through paracrine signaling and circulation throughout the body [30,31]. In adult tissues, elabela was found to be important mainly in the adult kidney, and could be detected in the plasma of pregnant, and it could inhibit kidney fibrosis and treat preeclampsia [8,32]. Moreover, Ela exhibits great potential in the treatment of cardiovascular diseases and can also suppress associated inflammatory responses [33]. Research indicates that Ela has a significant therapeutic effect on myocardial infarction, ischemia-reperfusion injury, heart failure, sinus arrhythmia, and primary and secondary pulmonary arterial hypertension [7,34–36]. Ela can alleviate cellular oxidative stress and inhibit iron-mediated cell apoptosis [37,38]. Furthermore, studies have demonstrated that Ela can treat doxorubicin-induced myocardial injury through the TFEB pathway, Akt/PI3K and ERK/-MAPK pathways [37,39]. All of the aforementioned research highlights Ela as a critical therapeutic and diagnostic target for cardiac diseases. We will explain the specific roles and molecular mechanisms in more detail in later sections.

2.3. Apelin receptor: APJ, function and signaling pathway

APJs is a typical GPCR consisting of a 7-transmembrane structure and coupled with G proteins. APJ is composed of 388 amino acids and its gene, APLNR, is located on chromosome 11 [3]. APJ expression is initiated when relevant transcription factors bind to APLNR. APJ is found to be widely expressed in the heart, blood vessels, kidneys, ovaries, and human early embryonic tissues [40]. Studies have shown that APLNR is expressed before gastrulation onset, indicating an indispensable role of APJ in early human embryonic development [24]. Severe embryonic developmental disorders and cardiac malformations were observed in mouse embryos with APLNR gene knockout [25,41]. Therefore, understanding the downstream signal transduction process of APJ is crucial to comprehend the causes of these diseases.

Currently known APJ ligands include the apelin family of peptides, as well as Elabela and Tollder recently discovered in zebrafish embryos and human embryonic stem cells [4,42]. The most obvious effect of APJ activation is the positive inotropic effect on myocardium and the contraction and dilation of the entire body's blood vessels [19,43,44]. However, it is noteworthy that the vascular effects mediated by apelin in vitro and in vivo are different [45,46]. For example, Hashimoto T et al.'s study found that directly administering apelin to mice led to a decrease in blood pressure, but applying apelin directly to arterial smooth muscle cells in vitro caused the arteries to contract. This suggests that the vascular effects mediated by apelin may not only be a single-cell effect but rather interact with multiple cells to exert their vascular effects [41].

Apelin can activate Gα/i to inhibit forskoline-stimulated production of cAMP and thereby affect downstream signal transduction [47]. Apelin can also activate Gα/q receptors to activate downstream Phospholipase C and protein kinase C, thereby exerting a positive inotropic effect and constricting blood vessels [48,49]. This positive inotropic effect can still have a therapeutic effect in the hearts of heart failure patients, indicating that apelin and related agonists developed based on apelin can be used as new therapeutic targets for heart failure treatment. Additionally, the vasodilatory effect of apelin depends on Gi-phosphorylated AKT to phosphorylate and activate eNOS[50](Fig. 2).

Meanwhile, studies have found that the short- and long-term stimulating effects of apelin or elabela can cause APJ to produce desensitization effects through β-arrestin, leading to GPCR internalization and ultimately reducing the stimulating effects of apelin and elabela [51]. If one wants to develop drugs targeting the apelin-APJ axis, the desensitization effects of GPCR cannot be ignored because they can lead to a reduction in drug efficacy.

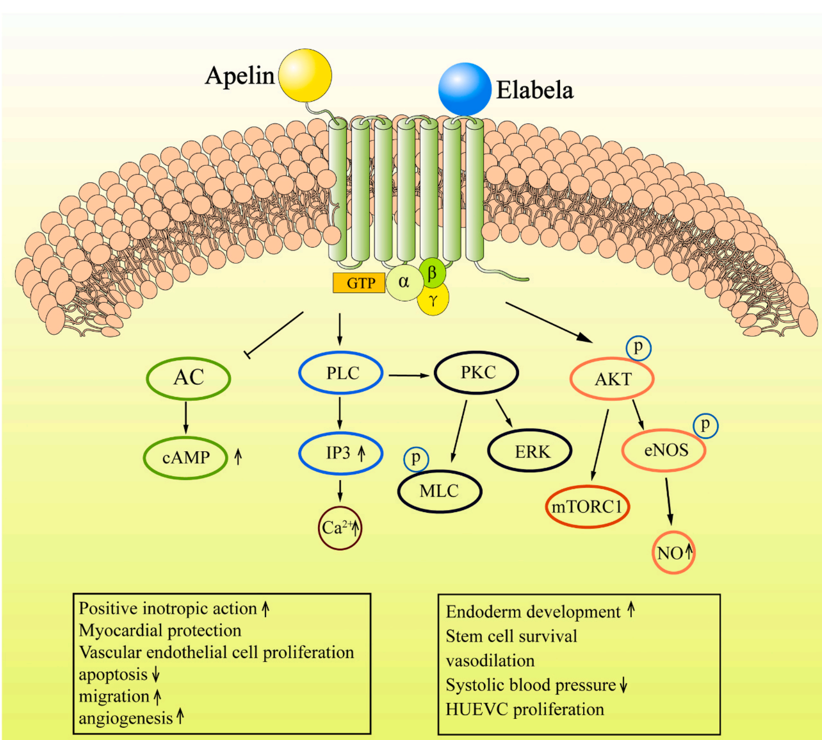

Fig. 2. Downstream signaling pathway of APJ. Apelin and Ela can bind to APJ and activate different subtypes of G proteins, triggering various downstream signaling pathways and resulting in diverse biological effects.
AC: adenylate cyclase; PLC: phospholipase C; IP3: inositol triphosphate; PKC: protein kinase C; MLC: myosin light-chain; ERK: extracellular signal-regulated kinase; AKT: protein kinase B; eNOS: nitric oxide synthase; NO: nitric oxide synthase; mTORC1: mammalian target of rapamycin 1.

3. APJ and downstream signaling pathway in heart disease

The binding of APJ ligand to APJ results in diverse effects, some of which play a crucial role in the treatment and regulation of cardiovascular diseases, particularly in diseases such as myocardial infarction, ischemia-reperfusion injury, pulmonary arterial hypertension, and atherosclerosis [8]. Therefore, it is essential to investigate how Ela and apelin affect the downstream signaling pathways of APJ and their impact on these diseases. Summarizing the roles of apelin and Ela in these diseases can provide us with a deeper understanding of the research and treatment of these conditions.

3.1. myocardial infarction and ischemia reperfusion injury

It is reported that about 8 million people die each year from acute myocardial infarction [52]. The features of myocardial infarction include the blockage of coronary arteries, leading to hypoxia in myocardial cells and the occurrence of anaerobic respiration [53,54]. At the same time, damage to mitochondrial function causes a large number of oxygen-free radicals to be released from the mitochondria, which damages myocardial cells and leads to severe oxidative stress [55]. However, restoring blood flow to the ischemic myocardium can potentially exacerbate the damage, a phenomenon known as ischemia-reperfusion injury [56]. During reperfusion injury, the inflammatory response is enhanced, and related inflammatory factors are released. The cell apoptosis signal pathway is over-activated, leading to more severe damage to myocardial cells, even death [56]. Meanwhile, the inflammatory response can cause extracellular matrix accumulation, leading to myocardial hypertrophy and ultimately heart failure after myocardial infarction [57,58].

Currently, many studies have shown that Ela and apelin can be used as targets for diagnosing and treating myocardial infarction and alleviating ischemia-reperfusion injury. Ela-32 treatment can activate the YAP/TAZ-TEAD pathway to regulate the expression of related pro-angiogenic factors CTGF and ANG2 to promote microvascular generation after ischemia-reperfusion injury [59]. Some studies have also shown that after Ela gene therapy, the expression of VEGF/VEGFR2 in mice increased, the Jagged1/Notch3 pathway was significantly upregulated, and mouse cardiac function was significantly improved, with increased angiogenesis [60]. G. Chen et.al found that the generation of these blood vessels may be through the apelin-APJ MAPK/ERK pathway to promote the survival of mesenchymal cells [61]. Oxidative stress and mitochondrial dysfunction are the main causes of ischemic myocardial cell damage, and Ela can protect myocardial mitochondria and tissue superoxide from mitochondrial damage. Additionally, it significantly promotes the effects of SOD and GSH, protecting myocardial cells from oxidative stress [34]. Ela can also promote the expression of bcl-2, thereby inhibiting the expression of cyt-c, a pro-apoptotic factor that can activate caspase3 and caspase9, leading to myocardial cell death, also bcl-2 can inhibit apoptotic molecular release from mitochondrial. The promoting effect of Ela on bcl-2 certainly mitigates ischemia-reperfusion injury in myocardial cells [62]. Ela and apelin also play an important role in compensatory blood vessel generation after myocardial ischemia, and apelin-13 can upregulate SDF-1α/CXCR-4 and related vascular progenitor cell homing mechanisms to promote vascular recovery after infarction and improve cardiac function [63]. Meanwhile, Ela also activates the ERK/HIF-1α/VEGF pathway to regulate post-ischemic vascular regeneration[34,64](Fig. 3). It is worth noting that relevant studies have found that after I/R injury, excessive extracellular matrix deposition often occurs, leading to heart failure after severe myocardial infarction [57,58]. However, treatment with Ela can significantly reduce the synthesis of extracellular matrix type I and III collagen fibers, thereby improving cardiac function after I/R [34,65]. Interestingly, there are also studies indicating that apelin-12, − 17, and − 36 can promote platelet aggregation through the pannexin1 (PANX1)-P2X7 signaling pathway [22]. It is still unknown whether this will exacerbate myocardial infarction caused by thrombosis or atherosclerosis-induced coronary artery occlusion, and more follow-up studies are needed to confirm this.

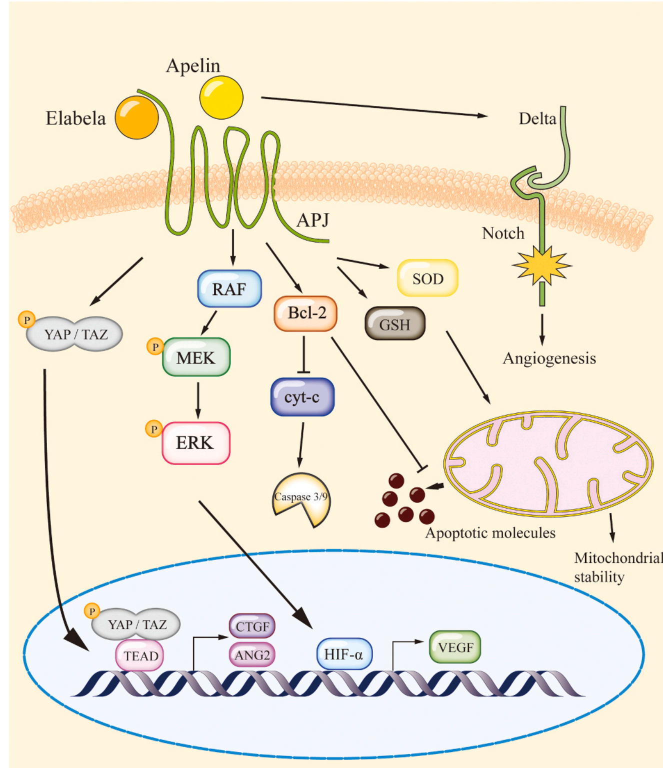

Fig. 3. Apelin, Elablea-APJ axis function during myocardial infarction. During ischemia-reperfusion injury and late-stage myocardial infarction repair, apelin and Elabela play crucial roles. Elabela can enhance the expression of GSH and SOD, promoting mitochondrial stability and protecting cardiomyocytes. Furthermore, they respectively activate the MAPK/ERK signaling pathway to facilitate VEGF production and activate the YAP/TAZ-TEAD pathway to induce the expression of ANG2 and CTGF, also enhance Delta/Notch signaling, thereby promoting post-myocardial infarction vascular regeneration. Additionally, Elabela can inhibit cardiomyocyte apoptosis by activating bcl-2. VEGF: Vascular Endothelial; RAF: RAF protein kinase; YAP: Yes associated protein; TAZ: Transcriptional co-activator with PDZ-binding motif; TEAD: Transcriptional enhancer associate domain; MAPK: Mitogen-activated protein kinase; ERK: Extracellular signal regulated kinase; BCL-2: B cell lymphoma 2; Cytochrome C; GSH: Glutathione; SOD: Superoxide dismutase; CTGF: Connective tissue growth factor; ANG2: Angiopoietin2; HIF-α: Hypoxia-inducible factor α; APJ: Apelin receptor.

3.2. Pulmonary arterial hypertension

Pulmonary arterial hypertension (PAH) is a devastating lung disease characterized by increased pulmonary vascular resistance, which is caused by pulmonary arterial endothelial damage, VSMC proliferation and fibrosis, as well as the formation of proliferative endothelial cell lesions [14,66–68]. This can ultimately lead to luminal obstruction, further elevating pulmonary pressure, resulting in right ventricular hypertrophy, and ultimately leading to right heart failure and even death [68]. The structural and functional changes of pulmonary artery smooth muscle cells (PASMCs), which are responsible for pulmonary vascular remodeling, are considered to be the main cellular mechanism of PAH [69,70]. Studies have found that the expression levels of apelin and Ela are significantly decreased in the plasma of PAH patients, as well as in the pulmonary arterial endothelium and pulmonary microvascular endothelium [51,71,72]. Furthermore, the expression of the overall apelin/Ela-APJ axis is also significantly decreased in the lung tissue of

PAH patients [73]. The level of apelin-17 in the serum or plasma of PAH patients can serve as a highly accurate diagnostic factor to assist in the diagnosis of PAH, and its diagnostic reliability can be comparable to that of established biomarkers such as NT-proBNP and GDF-15 [74].

Currently, several studies have shown that apelin and Ela can be used to improve and treat PAH. One of the important aspects of treatment is to rescue and reverse the apoptosis of endothelial cells and the remodeling of microvasculature. According to the study by Andrea LF et.al, estrogen 17β-estradiol (E2) can activate the BMPR2-dependent pathway through estrogen receptor alpha (ER-α) and increase the expression of downstream target gene apelin, thereby reducing the apoptosis of pulmonary arterial endothelial cells and the remodeling of pulmonary arterial smooth muscle cells [10]. In addition, apelin can also inhibit the proliferation and migration of hypoxia-induced mouse pulmonary arterial smooth muscle cells through the PI3K/AKT/mTOR pathway, preventing the excessive muscularization of pulmonary arterial vessels, and thus relieving right heart failure caused by PAH [75]. Meanwhile, apelin can activate nitric oxide synthase through Gα/i-AKT/eNOS, promoting the production of NO and relaxing the pulmonary arterial smooth muscle cells, thereby reducing pulmonary arterial pressure and relieving right heart failure [76].

Fan J et.al found that the expression of APJ was significantly reduced in PAH patients, which may be achieved through the regulation of NF-κB nuclear factor and MiR-335–3p function [77]. MiR-335–3p can directly act on its downstream target gene APLNR to cause damage to APJ expression, leading to the dysfunction of pulmonary arterial endothelial cells and smooth muscle cells [77]. Additionally, research has shown that miRNA is an important target for the treatment of PAH. The expression of miR-424 and miR-503 in pulmonary arterial endothelium was found to be decreased in PAH patients, resulting in a significant increase in the expression of FGFR1 and FGF2 directly targeted by miR-424 and miR-503, causing abnormal proliferation and lesion of pulmonary arterial endothelial cells. This pathway is closely related to the apelin-APJ axis, and the APLN-dependent miR-FGF axis can help maintain the stability of pulmonary vascular system [72].

Finally, due to the short half-life of apelin and Ela in the human body, their clinical applications in treatment still require more research.

3.3. Atherosclerosis

Atherosclerosis is a chronic inflammation of blood vessels characterized by the accumulation of low-density lipoprotein within the arterial wall, macrophage accumulation, foam cell formation after phagocytosis of lipoproteins, and proliferation of vascular smooth muscle cells, ultimately leading to the formation of atherosclerotic plaques [78–80].

Currently, there is limited research on Ela in atherosclerosis. What can be determined is that the level of Ela significantly decreases in the systemic circulation of patients with hypertension and increased intima-media thickness of the carotid artery [81]. Ela also has the effect of stabilizing the vascular endothelium and inhibiting the proliferation of vascular smooth muscle cells, which suggests that the downregulation of Ela expression may be an important factor contributing to atherosclerosis [82]. This requires further research for clarification.

The relationship between Apelin and the formation and progression of atherosclerosis is complex. Studies have shown that plasma levels of Apelin are closely related to the vulnerability of plaques and the incidence of acute coronary syndrome [83]. Apelin-13 can downregulate the expression of lipoprotein lipase in THP-1 macrophage foam cells through the APJ/PKCα/miR-361–5p signaling pathway. The overexpression of LPL accelerates the formation of foam cells and exacerbates the inflammatory response. Apelin-13 inhibits the accumulation of lipids and the release of inflammatory factors in macrophages through this pathway, thus alleviating atherosclerosis [5]. X. Y. Liu et.al also found through in vitro cell studies that Apelin-13 can activate PKCα to phosphorylate ATP-binding cassette A1 (ABCA1) and inhibit calcium-dependent proteinase-mediated protein hydrolysis, thus promoting cholesterol efflux and reducing foam cell formation [84,85]. Apelin-13 can also activate autophagy through the PI3K/Beclin-1 pathway, thus reducing lipid accumulation in foam cells. All of these findings suggest that Apelin can alleviate atherosclerosis by regulating macrophage lipid metabolism [6].

The apelin-APJ axis also has been shown to directly antagonize the cells and signal transduction associated with AngII, thereby inhibiting the development of atherosclerosis and abdominal aortic aneurysms induced by AngII in ApoE-/- mice [86]. However, it is worth considering that some studies have suggested that apelin may have a promoting effect on the formation of atherosclerotic plaques. Apelin-13 has been found to upregulate the expression of NOX4 and LC3II/I in ApoE-/- mouse models, which may lead to the accumulation of ROS in cells and promote the proliferation of VSMCs, accelerating the accumulation of lipids in plaques [87]. At the same time, apelin can also promote the expression of VCAM-1, thereby promoting the adhesion of monocytes to human umbilical vein endothelial cells. This process can also be completed by NOX-derived ROS, ultimately exacerbating atherosclerosis [88]. Furthermore, apelin can induce the migration of VSMCs through the PI3K/Akt/FoxO3a signaling pathway [89]. Whether this migratory effect will act on atherosclerotic plaques remains unknown and requires further research to prove.

4. Role of apelin and Ela in diseases during pregnancy and cardiovascular system development

4.1. Preeclampsia

Preeclampsia (PE) is a pregnancy-related hypertension disorder characterized by proteinuria and high blood pressure, with an overall incidence rate of 2–5%, posing a significant threat to the safety of both mother and baby [90]. Currently, the mainstream theory holds that abnormal function of trophoblasts and ineffective infiltration and invasion of the uterus, as well as ineffective remodeling of the uterine spiral arteries, are the main causes of PE [91–93]. Research has found that Ela expression is strictly zygotic and is highly expressed in the syncytiotrophoblasts (ST) layer and the primitive gut overlying the developing heart tube [8]. Furthermore, APJ has been found to be widely expressed in the trophoblasts and endometrium. This expression pattern suggests that Ela may regulate the function of cytotrophoblast and syncytiotrophoblasts in a paracrine manner to control the invasion of trophoblast cells into the maternal uterus. Researchers have found that compared with Ela+ /+ embryos, Ela-/- embryos have impaired angiogenic sprouting, vascular remodeling, and heart tube looping, even if blood vessels begin to form in Ela-/- embryos. This suggests that apelin and Ela play important roles in early embryonic development and the pathological process of preeclampsia [8].

Apelin and Ela are present in higher concentrations in maternal plasma, but lower in non-pregnant individuals. Ela concentration reaches its peak in mid-pregnancy, indicating its crucial role in early human embryonic development and maternal pregnancy [8]. Multiple studies have focused on using Ela and apelin to improve preeclampsia. H. Yang, X. Zhang et al. found that apelin can downregulate ferritinophagy pathway-mediated cell apoptosis in trophoblasts and that using Ela to treat erastin-treated mouse HTR-8/Svneo trophoblasts can rescue iron death caused by erastin [94]. Ela also successfully reversed hypertension, reduced trophoblast ROS, MDA, and iron ion levels, and rescued cell oxidative stress and mitochondrial damage [94]. Ela can also improve thinner placental labyrinth level, placental dysplasia, hypertension, and proteinuria problems in Ela-deficient mice [95]. Ela can also improve the survival and migration rate of human umbilical vein endothelial cells through the Ela-APJ axis and the PI3K/AKT pathway, indicating that Ela may improve ineffective remodeling of uterine spiral arteries in PE and thus treat preeclampsia [96]. The use of apelin can also improve histological abnormalities in PE. Researchers found that after treating PE mice with Pyr-apelin-13, the proteinuria of mice was significantly improved compared with that of the PE group, kidney function was restored, kidney damage was reduced, and oxidative stress markers 4-HNE and NOX-4 were downregulated in the tissue [97]. Additionally, researchers found that APLN expression was significantly reduced, and trophoblast migration was significantly impaired in trophoblasts treated with miR-193b-5p, suggesting that apelin is also an important target for the treatment of preeclampsia and improvement of kidney proteinuria [98].

4.2. Congenital disease in cardiovascular system

Congenital heart disease is the most common birth defect, affecting approximately 1.35 million newborns each year [99]. The Apelin/Ela APJ axis has been shown to play an important role in early embryonic development, heart development, and maintenance of human stem cell self-renewal [25,27]. As previously mentioned, due to the observed significant differences in embryonic development between APLN and APLNR knockout mice, Ela was predicted and subsequently discovered as the second ligand for APJ. Ela knockout results in embryonic lethality, with impaired heart development being the primary cause of death [25]. Ela null embryos exhibit severe cardiac defects, including the absence of a heart, a 95% decrease in the expression of the heart marker cmlc1 compared to normal embryos, and a significant mesendodermal lineage defect [25]. Research suggests that these cardiac defects and mesendodermal defects may be accomplished through their upstream transcription factors Gata 5 and Sox 17 [4]. Gata 5 is crucial for precardiac mesoderm to move toward the embryonic midline, thereby guiding CPC to move towards APLM and making the embryo capable of initiating heart formation [100]. At the same time, the lack of Ela leads to a decrease in the expression of Sox17, an important marker of the endodermal lineage that marks definitive endodermal precursors, which is crucial for guiding the overlying cardiac progenitors to the heart-forming region [4,101]. All of this suggests that Ela and Apelin are

Table 1
list of animal therapeutic experiments of apelin, Ela and their analogs.

Drug	Target of evaluation	Species	Dose	Signaling pathway	Effect	Reference
Pyr-apelin-13 (apelin)	cardiac muscle tissue	Wistar rats	10nmol/kg/d	Not mentioned	Protect myocardial cell infarction area ↓	[108]
Apelin-13 (apelin)	Pulmonary artery	male SD rats	10 nmol/kg/d	Not mentioned	Mean pulmonary artery pressure ↓ Right ventricular hypertrophy ↓	[109]
Apelin-13 (apelin)	Carotid vascular endothelial cells	ApoE-/- rat	2 mg/kg/d	Not mentioned	inflammatory response↓ Total cholesterol ↓ Low density lipoprotein ↓ Free fatty acid ↓	[110]
Apelin-13 (apelin)	System evaluation	albino rats	$2 \times 6 \times 10^{-8}$ mol/Kg/d	Not mentioned	Maintain uterine perfusion pressure	[111]
Pyr-apelin-13 (apelin)	Renal vessel	PE rats	2 mg/kg/d	Not mentioned	Blood pressure ↓ Ejection fraction ↑ Proteinuria ↓	[95]
[MeArg1, NLe10] apelin-12 (apelin)	Heart function evaluation	wenty maleChinchilla rabbits	2, 10 and 50 μg/kg/min for 10 min	Not mentioned	Ejection fraction ↑ Improve left ventricular systolic dysfunction	[112]
CMF-019 (Apelin analogue)	cardiac muscle tissue	Male Sprague–Dawley rats	Three cumulative doses 50–5000nmol	Gαi	heart contraction ↑ internalization of receptor↓	[113]
CMF-019 (apelin analogue)	Pulmonary artery	Sprague-Dawley rats	50–5000 nmol	Gαi	blood pressure↓ pulmonary artery endothelial cell apoptosis ↓	[114]
LIT01–196 (apelin analogue)	Blood pressure	Male WKYrats	15 nmol/kg intravenous infusion	eNOS	Blood pressure ↓	[115]
MM07 (Apelin analogue)	Pulmonary artery thickness, diameter Cardiac function	Male Sprague – Dawley rat	1 mg/kg/d intravenous injection for 21 days	eNOS/AMPK	Pulmonary artery endothelial cell proliferation ↑ Pulmonary artery vessels thickness and diameter ↓ Pulmonary artery remodeling↓	[76]
ELA-32 (Elabela)	Pulmonary artery	Male Sprague Dawley rats	450 μg/kg	Not mentioned	pulmonary artery remodeling↓ right cardiac hypertrophy↓	[36]
Ela-32 (Elabela)	Placental tissue	ELA-32 1 mg/kg/day	ELA-32 1 mg/kg/day	ferroptosis	Placental cell ferroptosis↓	[94]
Fc-ELA-21 (ELA analogue)	cardiac muscle tissue	Sprague-Dawley male rats	300 μg/kg	Not mentioned	Angiogenesis ↑ Myocardial cell proliferation ↑ apoptosis↓ myocardial fibrosis↓	[116]
BMS-986224 (APJ agonist)	hemodynamics	Male Sprague-Dawley rats	(1, 10, and 100 μg/kg/min) 15 min	cAMP/ERK	cardiac output ↑ Constant heart rate	[117]
6-hydroxy-5-phenyl sulfonylpyrimidin-4(1 H) (APJ agonist)	Acute hemodynamic effects	Male Sprague-Dawley (SD) rats	10 mg/kg/min (n = 8) via intravenous infusion over 15 min	Not mentioned	Cardiac Output ↑ Stroke Volume ↑	[118]
Hydroxypyrimidinone Compound (21) (APJ agonist)	Acute hemodynamic effects	Sprague-Dawley rats	1, 10, and 100 μg/kg/min × 15 min	Not mentioned	Cardiac Output ↑ Stroke Volume ↑	[119]

both critical for early cardiac and vascular development. More importantly, this also indicates that Ela and Apelin are likely to play important roles in early embryonic differentiation and related cell proliferation, migration, and division in conjunction with signal molecules such as Sonic Hedgehog, WNT, FGF, and others.

Recent studies have shown that Ela and Apelin levels in plasma are diagnostic markers for congenital cardiovascular diseases such as ventricular septal defects, pulmonary artery occlusion, and pulmonary artery stenosis [102]. Additionally, research has indicated that Apelin and Ela can mediate EMT and MET and thus play a pathological role in congenital heart disease [103]. Ela has also been found to act on Agtrl1b to mediate endodermal differentiation and heart development, and if this process is impaired, it may lead to congenital heart defects [13,104]. Although there is currently limited clinical research on Apelin and Ela in congenital heart disease, their roles in early heart development and trilaminar cell differentiation cannot be underestimated. Future studies on the signal transduction of Ela and Apelin in early embryonic development can provide new directions for clinical prevention, diagnosis, and treatment for congenital heart disease.

5. APELIN/ELA-APJ axis in cardiovascular system : therapeutic implication

As previously discussed, Ela and apelin play important roles in the development, pathology, and treatment of cardiovascular system-related diseases. In order to further understand the therapeutic effects of the apelin/Ela-APJ axis in cardiovascular disease, we summarize here the relevant effects of apelin, Ela, and their analogs in clinical treatment and animal experiments of cardiovascular diseases(Table 1).

In Table 2, we summarized recent clinical applications of apelin in the treatment of cardiovascular diseases. We found that the current applications of apelin and its analogs mainly focus on two areas. Firstly, the concentration of apelin in peripheral blood is used as a biomarker to monitor the treatment of diseases or as a target for prognosis and diagnosis in patients. Secondly, apelin can be utilized as a therapeutic agent for certain cardiovascular diseases such as hypertension and heart failure.

Researchers have indicated that injection of pyr-apelin-13 in patients with chronic stable heart failure effectively increases cardiac index and left ventricular ejection fraction. Compared to the placebo group, after one hour of pyr-apelin-13 injection, cardiac index and heart rate significantly increase, indicating a short-term improvement in cardiac function following pyr-apelin-13 administration [105]. There is also evidence showing that high-intensity interval training can lower blood pressure in middle-aged and elderly patients by upregulating apelin. After 6 weeks of high-intensity interval training performed three times a week, plasma levels of apelin and NOx increase, suggesting that high-intensity interval training can lower blood pressure and treat hypertension through the apelin-mediated eNOS pathway [106]. Another study found that aerobic exercise training (AT) can also increase apelin and NOx levels in hypertensive patients, thereby reducing blood pressure. These findings indicate that activation of the apelin-eNOS pathway may be one of the reasons why maintaining a good exercise habit can prevent the occurrence of cardiovascular diseases [107]. In conclusion, the apelin/Elabela-APJ axis remains a promising target for the treatment of cardiovascular diseases.

However, due to the short half-life and susceptibility to enzymatic degradation and lack of selectivity, there are currently few drugs targeting the apelin/Ela-APJ axis, and only AMG986 and a few other drugs have entered clinical trials. Currently, there are studies suggesting that modifying the C-terminal of drugs susceptible to ACE2 degradation is an important approach to enhance drug concentration and half-life in peripheral blood. Chemical or molecular modifications of the C-terminal amino acids can improve the pharmacokinetics of drugs, leading to greater therapeutic efficacy and improved treatment efficiency. The main focus of clinical drug development is biased activation of APJ and increasing the half-life [20]. Moreover, there are chemical substances emerging to address treatment tolerance caused by β-arrestin-mediated GPCR desensitization. Bias agonists are a favorable choice as they selectively target specific signaling pathways rather than fully activating all downstream signaling pathways of APJ, thus avoiding GPCR desensitization and potential side effects. Research has indicated that MM07 is an effective biased agonist of APJ. It selectively activates the G protein signaling pathway in the apelin-APJ axis, promotes proliferation of pulmonary vascular endothelial cells, increases cardiac output in rats, and does not diminish therapeutic efficacy upon repeated administration. This provides a proof-of-concept for the clinical potential of biased agonists in treating cardiovascular diseases through apelin and Elabela [124]. In the future，the most main focus of clinical drug development is on bias activation of APJ and increasing the half-life.

Table 2
Clinical trials using apelin as a therapeutic target.

Drug/ biomarker	Type	Result	reference
(Pyr(1)) apelin-13	Randomized Controlled Trial	peripheral resistance，mean arterial pressure↓ EF，CI↑	[105]
apelin	Controlled Clinical Trial	Apelin is a biomarker used to detect early endothelial dysfunction in these patients.	[14]
apelin	Randomized Controlled Trial	Blood pressure ↓	[106]
apelin	Randomized Controlled Trial	Peripheral and coronary artery dilation↑ CO↑	[120]
apelin	Clinical Trial	Blood pressure ↓	[121]
apelin	Clinical Trial	Apelin is biomarker for postoperative atrial fibrillation (POAF).	[122]
apelin	Randomized Controlled Trial	Patients treated with spironolactone might experience an increase in apelin levels in the blood, which can help prevent heart failure.	[123]
apelin	Randomized Controlled Trial	NO↑ Blood pressure↓	[107]

6. Conclusion and future perspectives

The research on the apelin/Ela-APJ axis is currently experiencing rapid expansion, with a growing number of clinical studies recognizing it as a promising diseases therapeutic target. Since 2013, the literature output in this field concerning apelin and Ela has reached approximately 2500 publications, with the cited references of these important studies exceeding 2000, and the publication rate continues to rise annually. This trend indicates that the apelin/Ela-APJ axis is increasingly gaining attention as a target for disease treatment, not only in the context of cardiovascular diseases but also in relation to cancer. Research has revealed that the apelin-APJ axis is associated with cancer, as it is upregulated in hypoxic tumor environments, promoting tumor metastasis, invasion, and dissemination through increased vascular growth [125]. The apelin/Ela-APJ axis is increasingly becoming a target for the diagnosis and treatment of various diseases, and related clinical experiments and therapies are actively being developed. In the future, further research in this therapeutic target is still needed to develop more efficient pharmacological tools and bias drugs, leading to a better understanding of disease treatment possibilities. Subsequent studies should focus on exploring how to utilize or modify apelin to achieve targeted delivery to specific tissues or selectively activate specific pathways in order to achieve a better therapeutic effect.

In conclusion apelin and Ela play a very important role in cardiovascular disease and are promising targets for clinical drug treatment. The summary of the knowledge of apelin and Ela in this article can provide a theoretical basis for further research on the regulation of cardiovascular development and the prevention of the above-mentioned pathologies.

CRediT authorship contribution statement

Shenghan Gao: Conceptualization, Writing – original draft, Writing – review & editing, Writing – review & editing, visualization. **Hongping Chen**: Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no conflict of interest.

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